Solution Problem 6.1

Signal model $x_k = f(x_{k-1}, w_{k-1}) = x_{k-1}^2 + w_{k-1}$, $E[w_k]=0$, $V[w_k]=\sigma_w^2$

Obs model $z_k = h(x_k, v_k) = x_k^3 + v_k$, $E[v_k]=0$, $V[v_k]=\sigma_v^2$

Jacobi-matrics

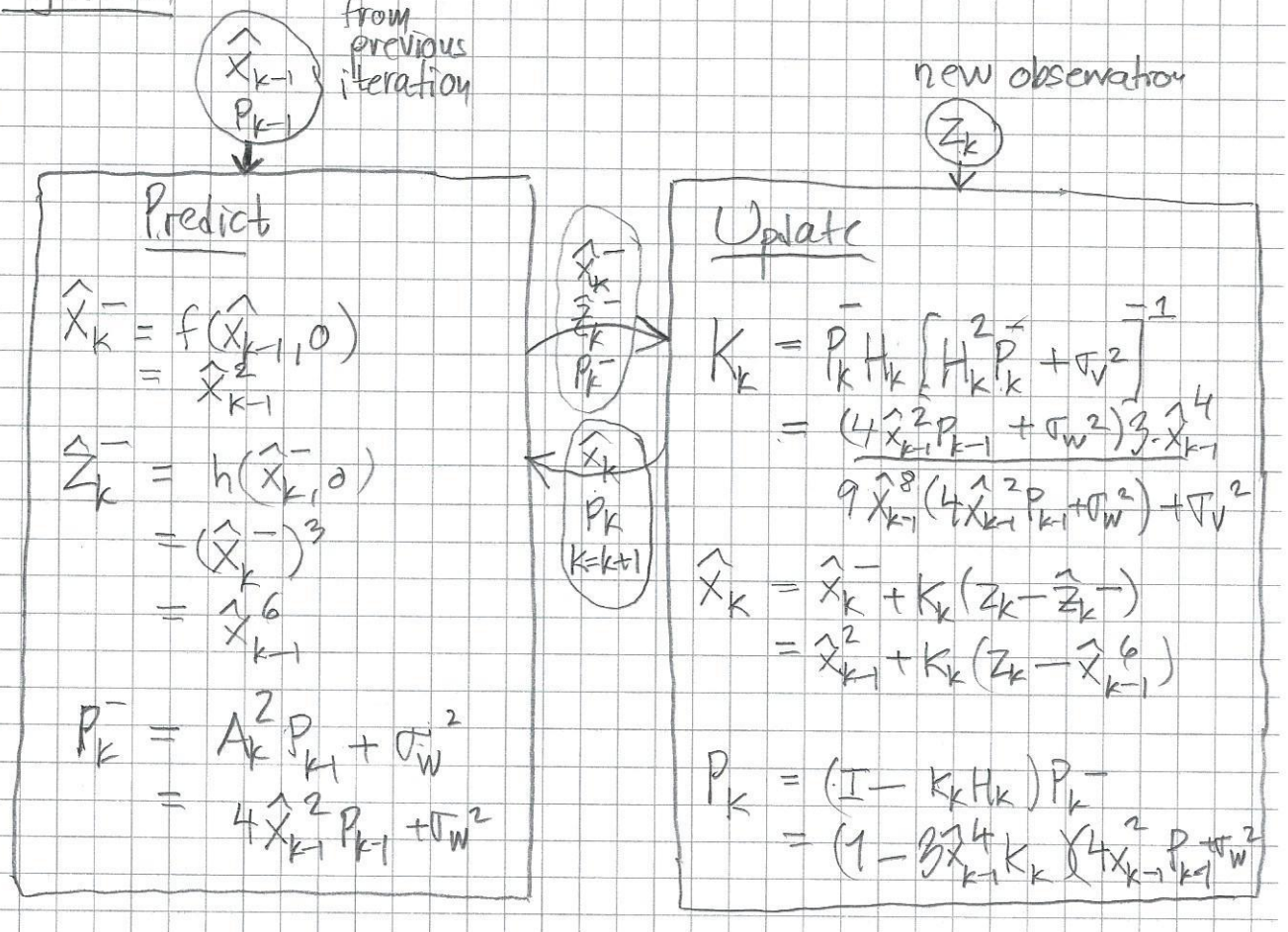
$$A_k = \left.\frac{\partial f}{\partial x}\right|_{\hat{x}_{k-1}} = 2\hat{x}_{k-1}$$

$$H_k = \left.\frac{\partial h}{\partial x}\right|_{\hat{x}_k^-} = 3\hat{x}_k^{-2}$$

$$W_k = \left.\frac{\partial f}{\partial w}\right|_{\hat{x}_{k-1}} = 1$$

$$V_k = \left.\frac{\partial h}{\partial v}\right|_{\hat{x}_k^-} = 1$$

EKF Algorithm



Solution Problem 6.2 (see also m-file)

CT SS

$$\frac{dx}{dt} = -\frac{K_v}{A}\sqrt{\rho g\, x(t) - p_0} + \frac{1}{A}u(t) + w(t)$$
$$z(t) = x(t) + v(t)$$

DT SS

$$x_k = x_{k-1} - \frac{K_v T_s}{A}\sqrt{\rho g\, x_{k-1} - p_0} + \frac{T_s}{A}u_{k-1} + w_{k-1}$$
$$z_k = x_k + v_k$$

Jacobians

$$A_k = \left.\frac{\partial f}{\partial x}\right|_{\hat{x}_{k-1}} = 1 - \frac{K_v T_s \rho \cdot g}{2A}\frac{1}{\sqrt{\rho g \hat{x}_{k-1} - p_0}}$$

$$W_k = \frac{\partial f}{\partial w} = 1$$
$$H_k = \frac{\partial h}{\partial x} = 1$$
$$V_k = \frac{\partial h}{\partial v} = 1$$

Covariances

$$Q_k = \sigma_w^2$$
$$R_k = \sigma_v^2$$

EKF algorithm

